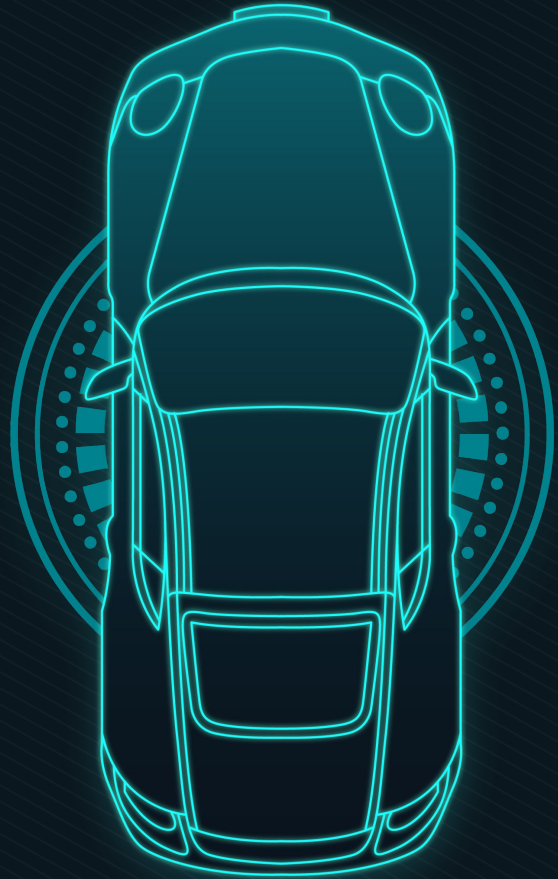


# FORMULA 1 RACE PREDICTION

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# CONCEPTUAL MODEL

- The conceptual model predicts each Formula 1 driver's race finish position using historical race, driver, and team data.
- Key input features include race-specific details (e.g., circuit, grid position), driver and team identifiers, and performance trends like average finish positions.
- The model uses supervised machine learning like **decision trees, random forests** to capture relationships and predict outcomes.
- Engineered features, like Top 3 finishes, enhance accuracy. Predicted outcomes support race strategies, sponsorship choices, and fan engagement.

# Business Problem



## Driver & Team performance Prediction

Anticipating which drivers or teams are likely to perform well in a given race, which could support race strategies, sponsorship decisions, and fan engagement.



## Race Strategy Optimization

Understanding performance trends could help teams optimize race strategies by identifying drivers who are likely to secure top positions, which is valuable for making decisions about qualifying sessions, pit stop timing, and tire management.



## Sponsorship and Marketing Insights

Brands and sponsors often prefer drivers and teams with high finishing probabilities, so predictions on race outcomes can guide sponsorships and partnerships.



## Fan Engagement and Betting Markets

Predictive insights can enhance fan engagement, particularly for fantasy sports leagues or betting markets, where outcome probabilities play a significant role.

# Machine Learning Problem

- **Objective:** Predict the finishing position of Formula 1 drivers in a race based on historical race data, including starting positions, driver information, and constructor details.
- **Approach:** Use machine learning models (e.g., Random Forest Classifier) to classify drivers' finishing positions, with evaluation based on accuracy.
- **Dataset:** Utilize data on races, drivers, constructors, and results, capturing features such as starting grid position, team performance, and race status.

# TYPE OF LEARNING

- **Type of Learning:** This is a **supervised learning problem** as the model is trained on historical data with labeled outcomes (`positionOrder`), indicating drivers' finishing positions.
- **Features and Labels:** Input features include race details (`raceId`, `grid position`, `laps`), driver information (`driverId`, `constructorId`), and performance metrics (`fastestLapTime`, `fastestLapSpeed`), while the target label is the finishing position (`positionOrder`).
- **Goal:** The objective is to predict drivers' finishing positions in future races based on historical data and observed patterns.

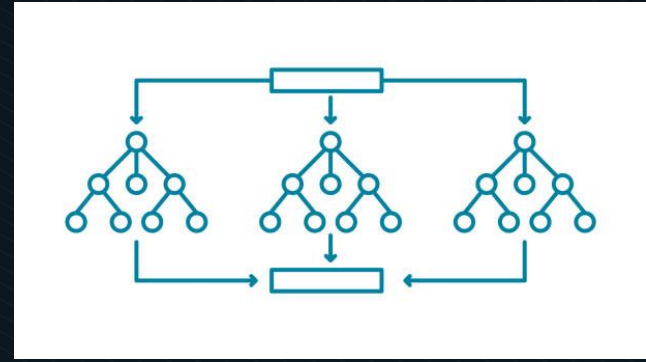
# ML MODEL

## Model Type: Decision Tree Classifier

- A supervised learning algorithm used for **classification** tasks.
- Predicts outcomes (e.g., finishing positions - Top 3 finishes) based on historical race data.

## Advantages:

- Handles both categorical (Driver/Constructor IDs) and numerical (Grid Position) data.
- Provides interpretable results via feature importance analysis.





# SAMPLE DATA

- **Drivers Dataset:** Details about F1 drivers, including their IDs, names, and nationalities.
- **Constructors Dataset:** Information about F1 constructors (teams).
- **Races Dataset:** Data about F1 races across years.
- **Results Dataset:** Results for each driver in every race.
- **Circuits Dataset:** Information about the circuits used in F1 races.
- **Lap Times Dataset:** Detailed lap-by-lap performance of drivers during each race.





# TARGET VARIABLE

## Target Variable Setup



### 1. Podium Prediction (`is_podium_finish`)

- **Definition:** Predict whether a driver finishes in the top three (1 for yes, 0 for no).
- **Setup Source:** Generated from the `positionOrder` (finishing position) column in the dataset.
- **Application Scenarios:**
  - Strategy development for racing teams.
  - Precise odds setting for the betting industry.
  - Providing key race highlights to the audience.

### 2. Finishing Position Prediction (`finishing_position`)

- **Definition:** Predict the specific finishing position of a driver (1 for first place, 2 for second, and so on).
- **Setup Source:** Directly use the `positionOrder` column in the dataset as a multi-class target.
- **Application Scenarios:**
  - Race simulation and driver performance analysis for race operators.
  - Optimized resource allocation and strategy development for racing teams.

# TARGET VARIABLE

## Value of Target Variables

- **Accurate Predictions:** Enhance the predictability of race results, supporting the diverse needs of teams and audiences.
- **Strategic Decision-Making:** Help racing teams design more optimized race strategies, boosting competitiveness.
- **Commercial Value:** Provide data-driven insights for betting and marketing purposes.

## Extended Target Variables

We can also explore other potential target variables:

1. **Did Not Finish Label:** Predict whether a driver will retire from the race.
2. **Completion Time Gap:** Quantify the time difference between a driver and the race winner.
3. **Team Ranking:** Predict the overall performance of an entire team.



# USEFUL FEATURES



1. **Racer:** Drivers skill levels and experience are major determinants of race outcomes.
2. **Team (Constructor):** The team's performance and the car's engineering significantly affect outcomes.
3. **Circuit:** Each F1 circuit is unique such as track layout, weather conditions, and altitude, which impact race outcomes.
4. **Qualifying Position:** Starting position greatly impacts the chances of a podium finish.
5. **Weather Conditions:** Weather plays a crucial role, as rainy conditions affect race strategies and the likelihood of accidents.
6. **Pit Stops and Tire Choices:** The number of pit stops and tire strategies employed by each team often impact race results

# BUSINESS VALUE



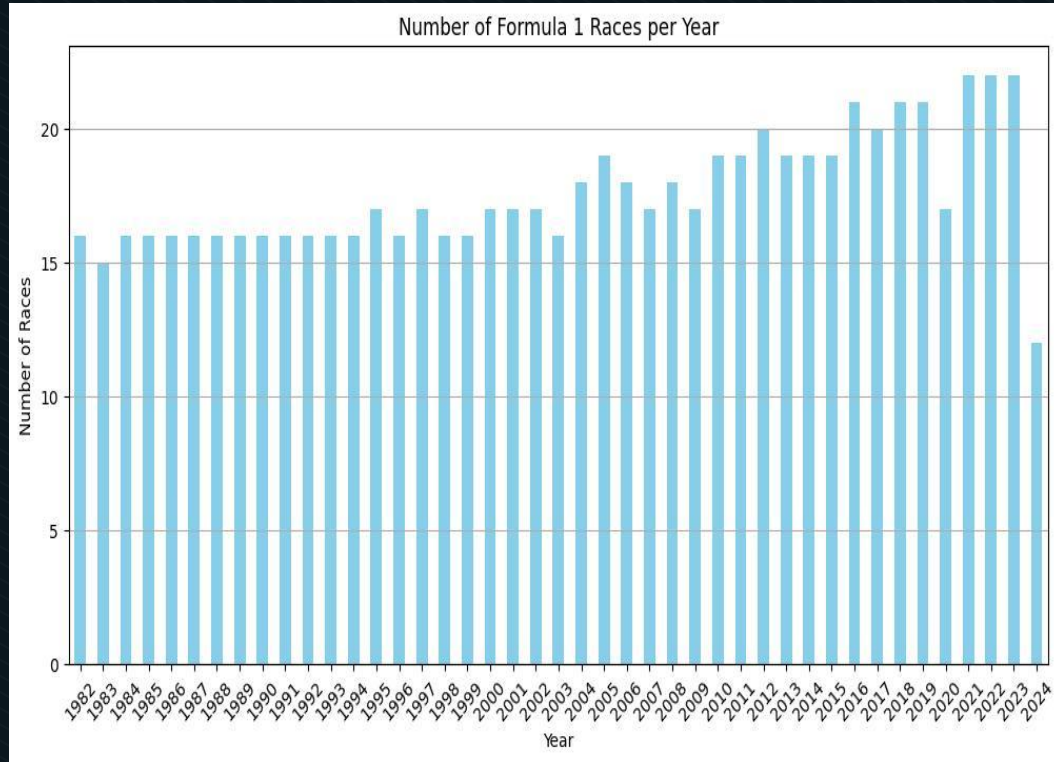
**Fan Engagement:** Accurate predictions boost fan interest, driving revenue for sports companies.

**Team Strategy Optimization:** Teams use predictive insights to make data-driven race decisions, increasing podium chances.

**Sponsorship and Advertising:** Targeting high-performing drivers and teams helps attract sponsorships, raising revenue.

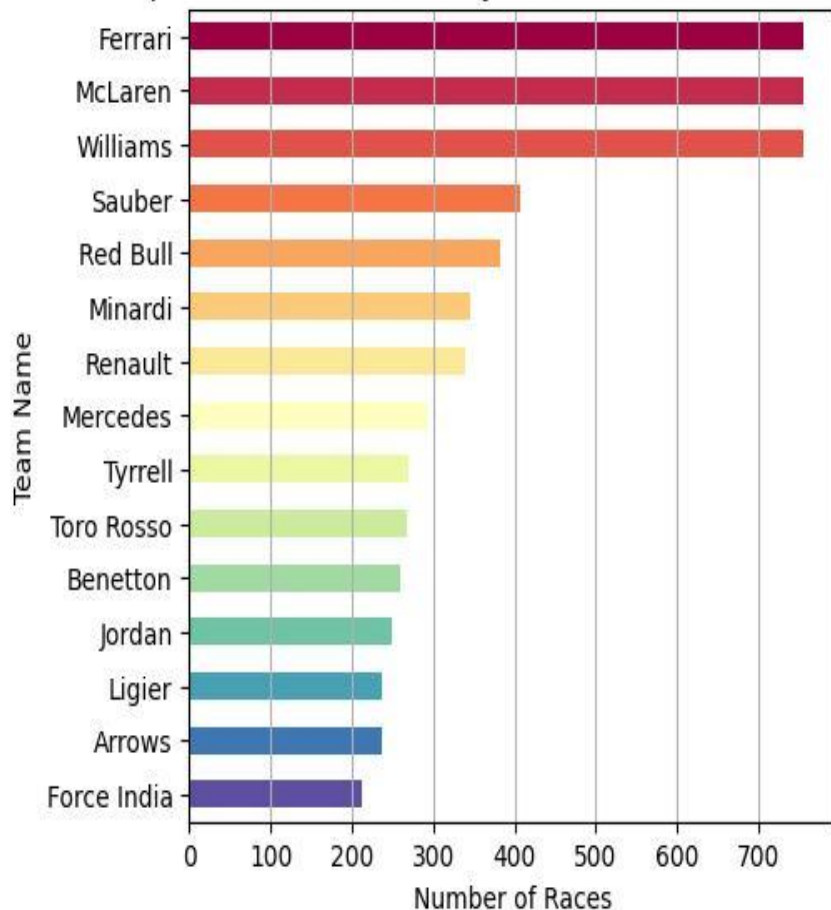
**Media and Broadcasting:** Media can create engaging, prediction-based content to attract viewers and increase ad revenue.

# Data Graphs



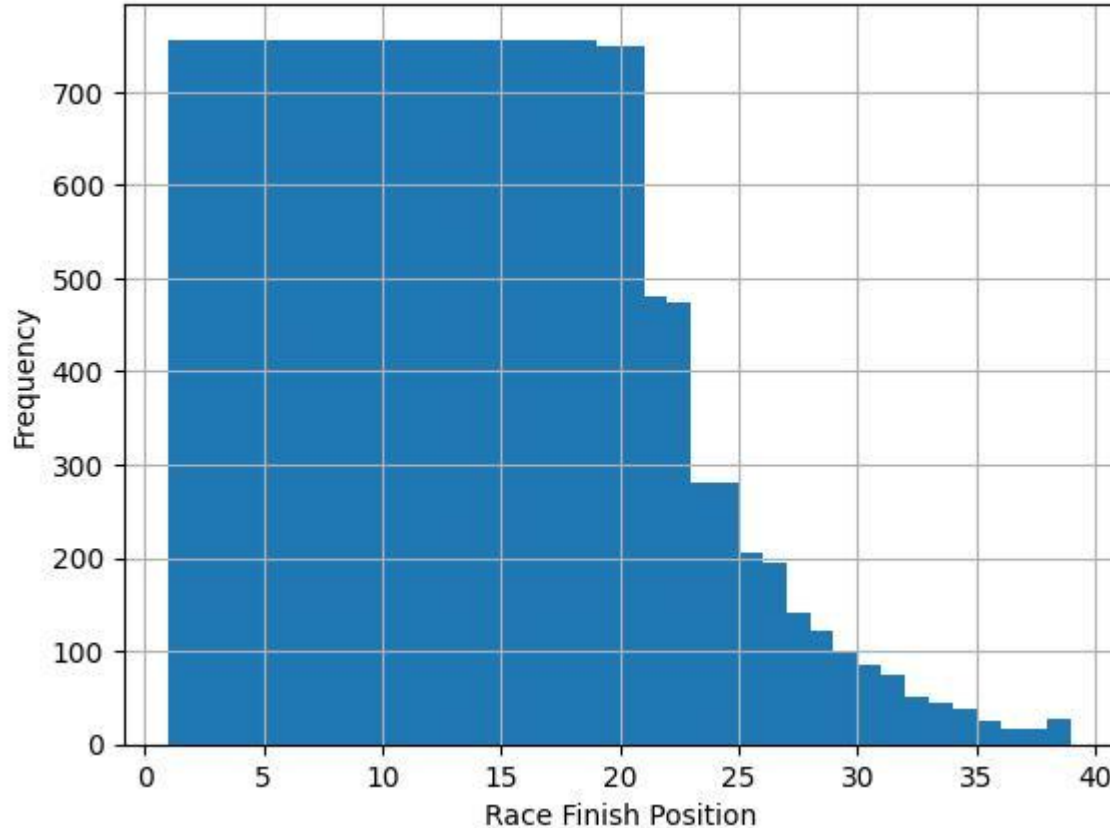
1. The number of races remains relatively consistent at around 15-16 races per year during the 1980s and 1990s.
2. By the 2010s, the calendar includes over 20 races per season.
3. The dip in 2020 likely corresponds to the COVID-19 pandemic.
4. The trend resumes its upward trajectory post-2020.

Top 15 Formula 1 Teams by Number of Races since 1982



1. **Ferrari, McLaren, and Williams** lead the chart, showcasing their long-standing presence.
2. Teams like **Sauber, Red Bull, and Renault** follow.
3. Newer teams like **Mercedes**, despite a shorter timeline, have managed to establish themselves prominently.
4. Lower-ranked teams, such as **Force India** and **Arrows**, have participated in fewer races, often due to shorter stints in F1 or rebranding over time.

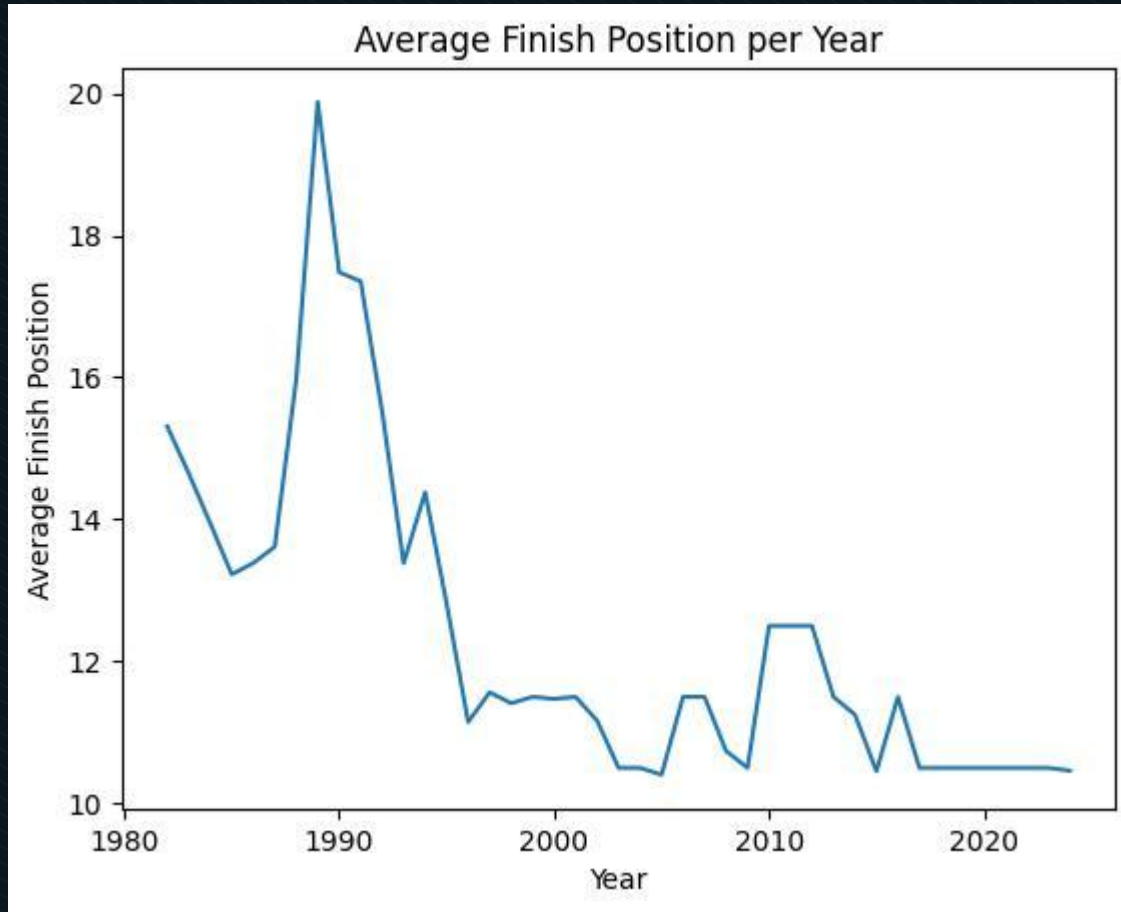
Distribution of Race Finish Positions



## Insights from the Distribution of Race Finish Positions

1. The distribution is right-skewed, with most races finishing in top positions (1–10).
2. Positions 1–5 have the highest frequency, reflecting strong performance by leading drivers or teams.
3. The frequency gradually declines beyond position 10, showing fewer consistent mid-pack finishes.
4. A long tail extends up to position 40, likely representing rare poor finishes or DNFs (Did Not Finish).
5. The data highlights the competitive importance of achieving top-10 finishes.
6. Teams should prioritize strategies to consistently secure top finishes to maximize performance impact.





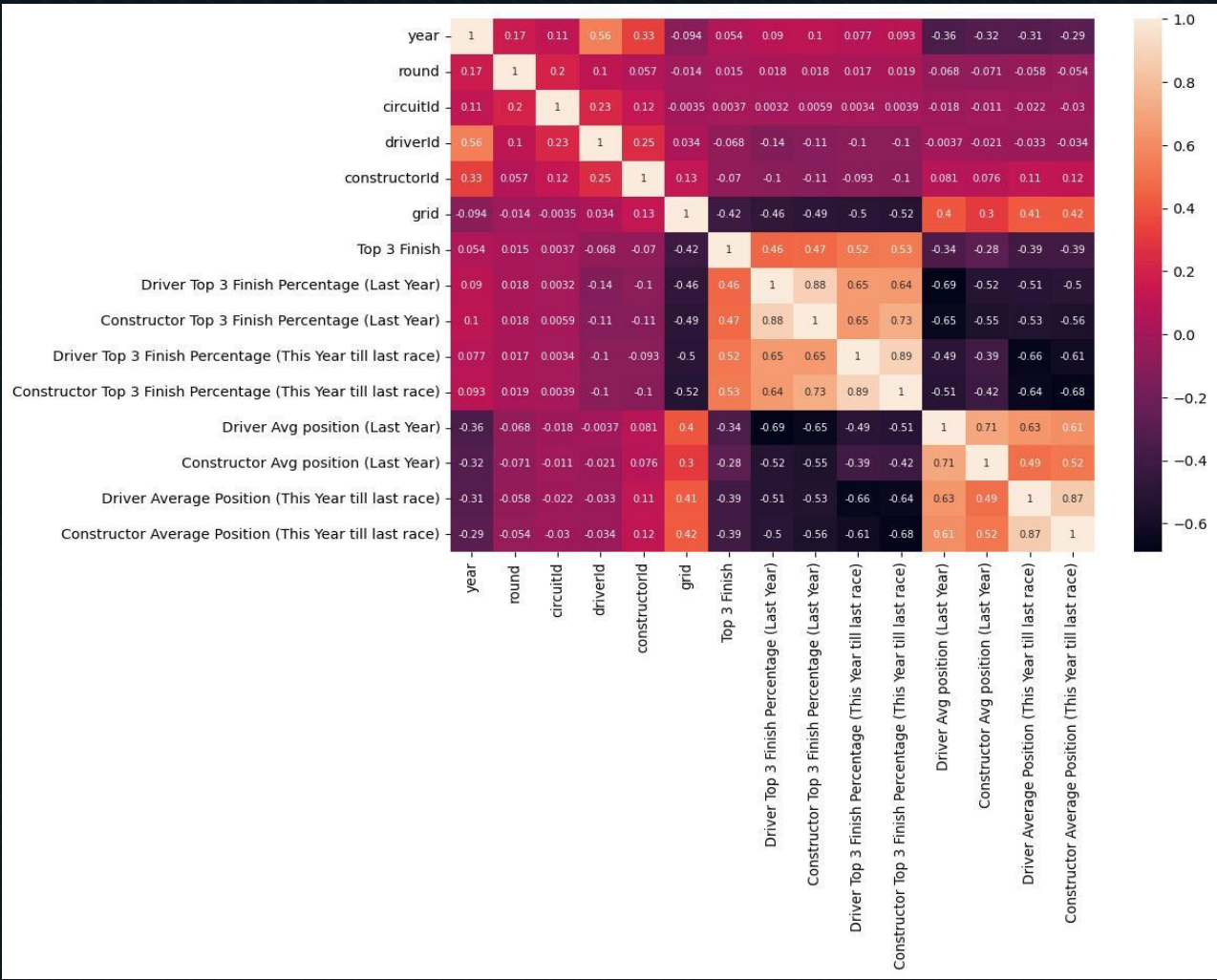
## Trend in Average Finish Position per Year

1. The chart shows the average finish position trends from 1980 to 2020.
2. A sharp peak in the late 1980s to early 1990s indicates performance volatility during that period.
3. A steady improvement is observed from the 1990s to the early 2000s, reflecting better consistency.
4. Since the mid-2000s, the average finish position has stabilized around 10–12.
5. The trend suggests improved competition balance and regulation impact in recent years.
6. This analysis highlights the evolution of performance and dominance in motorsport.

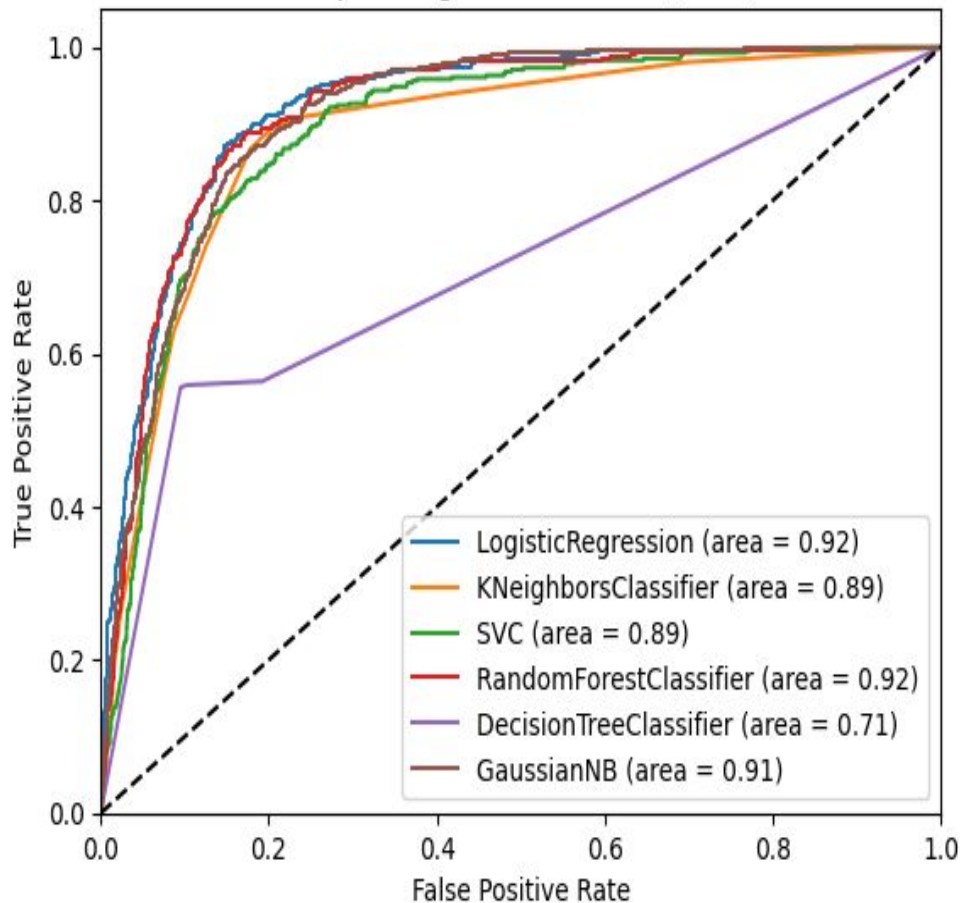
# Output Graphs

# Correlation matrix

- 1. Highlights correlations among key racing metrics.
- 2. Strong positive correlation observed between top 3 finishes and performance percentages.
- 3. Negative correlation between average position and top finishes indicates better rankings improve results.
- 4. Constructor and driver performance metrics are interlinked.



## Receiver Operating Characteristic (ROC) Curves



**Key Metric:** The Area Under the Curve (AUC) is displayed for each classifier. A **higher AUC indicates better model** performance in distinguishing between classes.

**Best Performers:** **Logistic Regression** and **Random Forest Classifier** both achieve an AUC of 0.92, indicating they are the top-performing models in this comparison.

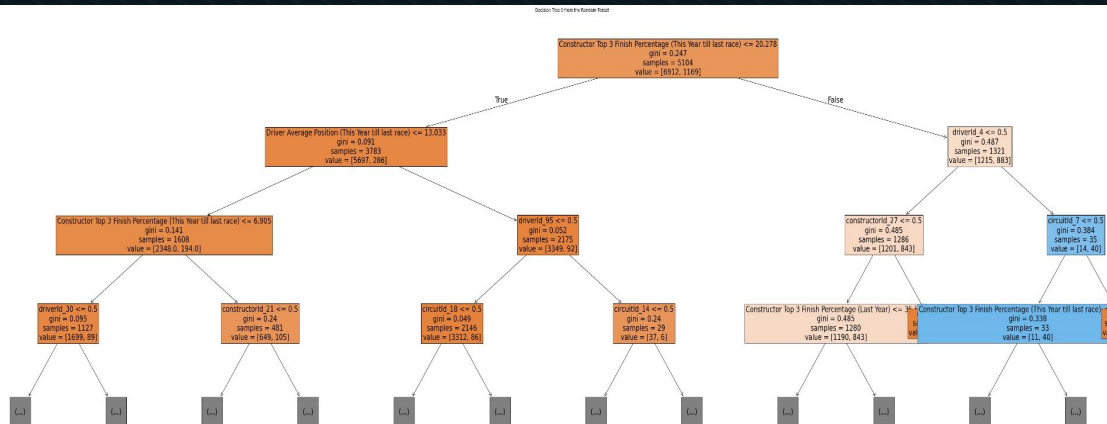
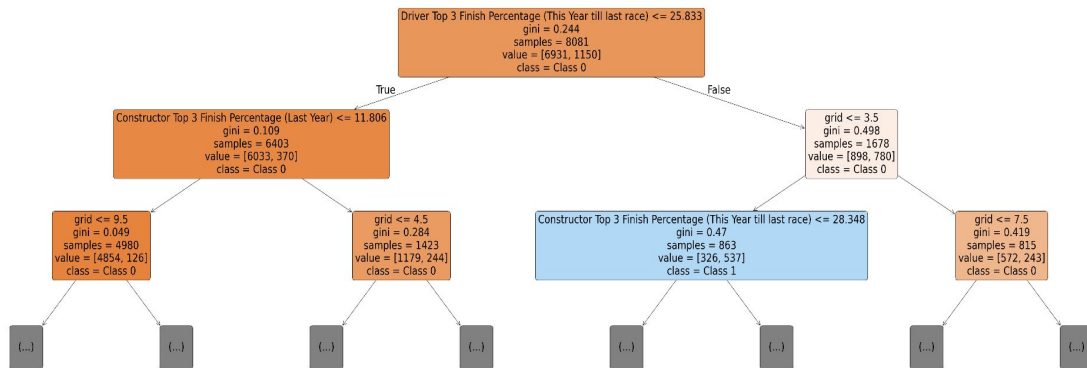
**Other Strong Performers:** **Gaussian Naive Bayes** has an AUC of 0.91, close to the best models. **K-Nearest Neighbors (KNN)** and **Support Vector Classifier (SVC)** both have an AUC of 0.89, slightly lower than the top models but still competitive.

**Weaker Performer:** The **Decision Tree Classifier** has the lowest AUC (0.71), showing it struggles compared to the others in distinguishing between classes effectively.

### ROC Curve Shape:

- A steeper curve (closer to the top-left corner) indicates better performance, with fewer false positives and higher true positives. Logistic Regression and Random Forest curves dominate the top-left corner, reaffirming their superiority.

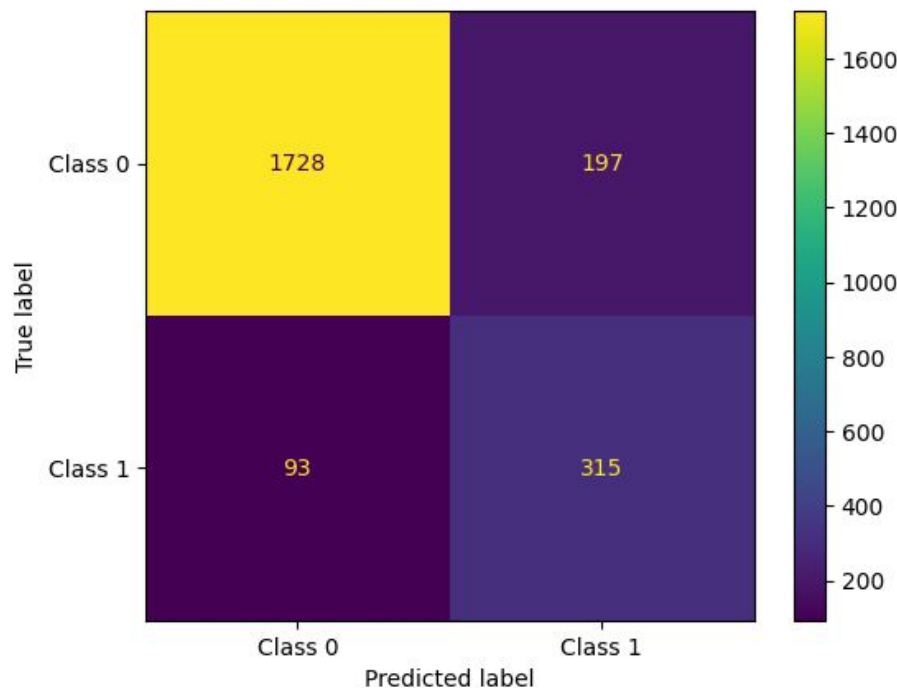
# Decision Tree



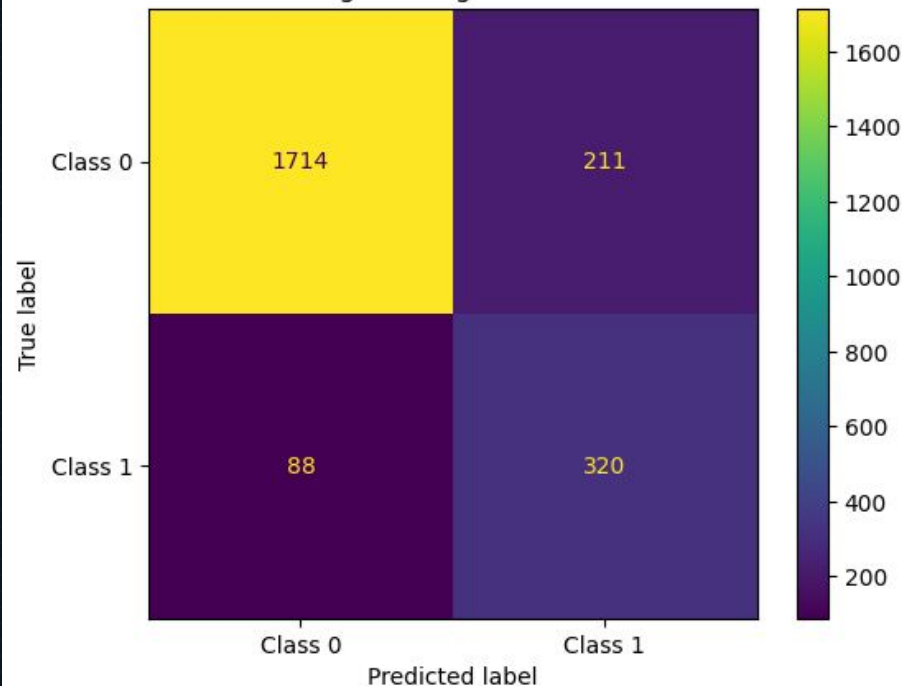
- The decision trees visualize the **key factors influencing race outcomes**, highlighting features like **Driver/Constructor Top 3 Finish Percentages, Grid Position, and Driver Average Position**.
- Splits are driven by metrics like **Gini impurity**, with historical performance emerging as the strongest predictor.
- These insights guide **targeted strategies** for improving race predictions and performance analysis.

# Correlation matrix

Confusion Matrix for Random Forest Classifier on Test Data



Confusion Matrix for Logistic Regression Classifier on Test Data



# Key takeaways Correlation matrix

## Performance Comparison:

- Both models perform well, but **Random Forest** has slightly **better precision for Class 0** (fewer false positives).
- **Logistic Regression** performs marginally **better for Class 1**, with fewer false negatives.

## Overall Prediction Accuracy:

- The models are closely matched, with Random Forest being more conservative and Logistic Regression slightly better at identifying Class 1 instances.

## Use Case Implication:

- To **minimize false negatives** (Class 1 misclassification) is crucial, choose Logistic Regression.
- To **reducing false positives** (Class 0 misclassification) is a priority, Random Forest is preferable.



# Conclusion

## 1. Podium Prediction Model

- **Objective:** Predict whether a driver will finish in the top 3 positions.
- **Type of Model:** Classification model (binary classification for each driver: podium finish vs. not podium finish).
- **Features:**
  - Driver-specific data: historical performance, current season form, qualifying position.
  - Team data: car performance metrics, constructor rankings.
  - Track data: circuit type, weather conditions.
  - Race-specific data: starting grid position, penalties, pit stop history.
- **Output:** Probability of a podium finish (custom threshold, e.g., 0.369, converts probabilities into binary decisions).
- **Evaluation Metric:** F1 score to balance precision and recall, as imbalanced classes may exist (few podium finishes compared to non-podium finishes).

## 2. Driver Finishing Position Prediction Model

- **Objective:** Predict the exact finishing position of each driver in a race.
- **Type of Model:** Multi-class classification model (predicting ranks for positions, e.g., 1st, 2nd, 3rd, etc.).
- **Features:**
  - The same as the podium prediction model, but potentially enriched with sequential or time-series data capturing race progression.
- **Output:** A rank (integer) representing the predicted finishing position.
- **Evaluation Metric:** Accuracy, measuring how often the predicted position matches the actual finishing position.

# Thank You

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## FOMULA

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## 2.3

THANK YOU

— OLEFANZ — VOX